

The Vector Is the Answer?

One-Pass Residual Readouts vs Fair Loop Scoring

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Abstract

Language models usually answer classification questions by generating an answer token, but their internal representation may already contain the information needed to choose the correct answer. We test this by comparing two ways of answering questions with a fixed choice set. The first is a small classifier on the frozen residual after one forward pass. The second is fair closed-set scoring: the model sees the question and balanced few-shot examples, and we compare answer-token probabilities rather than free-form or multi-step generation. The classifier is trained on the gold training split; the loop is few-shot scoring, not a fine-tuned decoder.

We evaluate on BoolQ, RuleTaker, and ARC-Challenge with six open-weight models from 0.6B to a mixture-of-experts model. The one-pass classifier matches or beats token scoring on BoolQ and RuleTaker except two cells (Mistral-7B on BoolQ; DeepSeek-V4-Flash on RuleTaker). On ARC-Challenge the scorer wins when the one-pass classifier is still weak — that tracks probe accuracy, not model size — including an 8.2-point gap on Granite-3.1-8B. The classifier’s largest gain is 11.9 points on RuleTaker with Granite-3.1-8B, where token probabilities are weakest. Label-shuffle, random-projection, and paired tests support the comparison. Sweeping the tap across depth, the best mid-layer readout beats the final layer in 10 of 18 cells after correction and never loses significantly. That is what a small readout can extract, not a license to skip later layers. For this task class the vector often already contains the answer; the loop still wins on ARC when the probe is weak.

1 Introduction

Autoregressive decoding is the usual interface to a language model, but it is only one readout of a residual vector that already exists after a single forward pass. On closed-set classification, does that vector already contain the gold label?

We compare two readouts of the same frozen model. The *one-pass readout* is a small classifier — logistic regression or a one-hidden-layer MLP — fit to the final-token residual after one prompt pass; the backbone is never updated. Linear probes sit within a few points of the MLP (§3). The *fair loop* is closed-set scoring: the model sees the question plus balanced few-shot exemplars, and we compare next-token log-probabilities over the answer set. Zero-shot prompting understates this baseline [8]. A probe that recovers the model’s self-report shows beliefs are present, not that a readout matches a fair loop at the gold label.

The loop is given balanced few-shot exemplars and a longer context than the readout (deliberately; §2). The readout is fit on the training split. Extra in-context shots do not erase the pattern on BoolQ (§3). We report label-shuffle nulls, random-projection selectivity, and paired tests.

Concurrent work. Hazenoot et al. [20] find a linear probe on frozen activations beats the same model’s prompted yes/no on ESG concept measurement (11/12 comparisons), mid-network. Their baseline is zero-shot prompting, which understates the loop (§3), and they report no paired significance or selectivity controls.

Findings. BoolQ (full validation, 3270 rows), RuleTaker (10k/4k), ARC-Challenge (full test); six models from 0.6B to a frontier MoE:

1. **BoolQ and RuleTaker: match or win, with two exceptions.** The one-pass readout matches or beats the fair scorer except Mistral-7B on BoolQ (-2.2pt) and DeepSeek-V4 on RuleTaker (-1.6pt). Four of six BoolQ cells are statistical ties within $\pm 1.2\text{pt}$ (§3, §4).
2. **On ARC the loop wins when the probe is weak.** The gap tracks probe accuracy, not parameter count: Qwen $-10.6 \rightarrow -4.1 \rightarrow -0.4\text{pt}$ (the last ns); Granite-8B still loses by 8.2pt at readout accuracy 0.708 (§5).
3. **Answer-relevant information is already present mid-depth.** The best mid-layer tap beats the final layer at 11/18 cells (10/18 after Benjamini–Hochberg) and never loses. That is what a small readout can extract from h_l , not a claim that later layers can be skipped (§4.4).
4. **The margins can be large.** Where token probabilities are weakest, the readout’s lead is largest: $+11.9\text{pt}$ on RuleTaker with Granite-3.1-8B (readout 0.823 vs loop 0.703 , McNemar $p = 1.1\text{e}-45$).

Fine-tuned specialist encoders beat our readout; serial-depth tasks may require the loop; a frozen latent loop drifts (§5, §6). The result is for this task class, not early exit and not all tasks.

Scope. A companion position paper explores a serving architecture of kilobyte-scale readout heads (Paper 2). This paper contains no architecture claims and no production system.

2 Setup and Method

2.1 Two readouts of the same vector

One-pass readout. Let $h_T \in \mathbb{R}^d$ be the final-token residual of a frozen model after one forward pass over the task prompt (question + passage, truncated at `pad_max = 384`). We fit a head $f(h_T) \rightarrow \hat{y}$: either multinomial logistic regression or a one-hidden-layer MLP (width 128, 4-seed vote), both on standardized features. The head sees only task labels; the backbone is never updated.

Fair loop. The autoregressive baseline is *not* greedy decode — it is the strongest *classifier* the loop provides. The model receives the full context (balanced few-shot exemplars, `loop_pad_max` 2048–8192) and is scored by next-token log-probability over the closed answer set (Yes/No, or A–D). We report `loop.0` (zero-shot) and `loop.k` (k -shot); where noted we sweep k up to 64 with a $4\times$ pad budget. *Scope:* we evaluate the strongest *classifier* the loop offers, not a chain-of-thought loop that generates intermediate tokens before answering. Generated-then-answered chains are a different readout, break the matched closed-set scoring protocol, and — on a serial task such as RuleTaker — require intermediate generation by construction; they are out of scope for the matched comparison.

2.2 Matching requirements

A readout-vs-loop comparison is only meaningful if the loop is not handicapped. We match **scoring** (same closed answer set, accuracy on identical rows). On **information** we advantage the loop: both see the same question and passage, and the loop also sees k balanced exemplars from the training split plus a longer pad (2048–8192 vs. the readout’s 384). That is conservative for the readout: the loop sees more of the passage and the exemplars. The readout is a probe fit on the training labels; the loop is in-context scoring. We sweep the loop to $k \leq 64$ at $4\times$ pad on BoolQ at 0.6B/4B/8B; extra shots plateau and do not pass the readout where it was winning (§3).

2.3 Controls

Every headline number carries four controls:

- **Label-shuffle null:** the head refit on permuted labels must sit at chance, ruling out leakage and spurious separability.
- **Random-projection selectivity:** a head on random Gaussian projections of the residual. If it retains accuracy, the signal is *diffuse* — no privileged subspace — which is what we find (permuted-basis $\approx$ max-basis $\gg$ noise floor) at every scale.
- **Final-token vs full-context:** a mean-pooled full-context readout must underperform the final-token readout, confirming the verdict is *computed into* the final state rather than read off the passage surface.
- **Paired significance:** McNemar’s exact test and paired bootstrap on per-row predictions from identical val rows, not pooled-accuracy eyeballing.

On BoolQ at DeepSeek-V4 we also re-score within question- and passage-length terciles (accuracy 0.89–0.90, flat). That is a coarse check that the readout’s edge is not an artifact of the loop truncating long rows; it is not a control on every cell.

2.4 Models and artifacts

Six open-weight models spanning three families and a width ladder: Qwen3-0.6B (28×1024), Qwen3-4B (36×2560), Qwen3-8B (36×4096), Mistral-7B (32×4096), Granite-3.1-8B (40×4096), DeepSeek-V4-Flash (43×4096 , FP8 MoE). All runs are batch 8 on model-pinned GPUs (0.6B/4B H200, 8B/Mistral/Granite B200, DSV4 B300). Trained heads are persisted per layer as versioned artifacts (~ 0.5 – 8.4 MB each); per-row predictions persist for paired tests. Run manifests embed the batch tag in cache keys.

3 BoolQ: one-pass readout vs the fair loop

BoolQ [10] is binary reading comprehension with the gold answer grounded in a passage — the cleanest closed-set case in which to test whether one prompt forward pass already carries the label. We load the standard Hugging Face release (google/boolq) and use the full validation split (9427 train / 3270 val).

3.1 Headline

Across the Qwen and Granite scales, the one-pass readout **wins at 0.6B, is a statistical tie at 4B/8B (numerical losses ~ 0.8 – 1.2 pt, ns), and trails the few-shot loop by ~ 1 pt (ns) at Granite-8B and the frontier MoE** (Table 1). The one exception is Mistral-7B, where the loop wins by 2.2pt ($p = 2.6e-03$) — the single significant loop win on BoolQ, which we flag as a single-model exception (§6).

3.2 Extra in-context shots do not close the gap

The natural objection: the head sees thousands of labels, the loop sees k exemplars. We gave the loop its budget back — k swept to 64 balanced exemplars at $4 \times \text{pad}$ (8192) — at 0.6B/4B/8B. The loop’s k -curve **plateaus by $k=32$** (loop.64 – loop.32 = -0.0003 , $p = 1.00$ at 0.6B) and **never passes the readout where the readout was winning:** at 0.6B the readout’s $+3.0$ pt over the best loop stays significant ($p = 3.2e-05$); at 4B/8B all cells are exact ties. More in-context supervision does not explain the 0.6B win (Fig. 2).

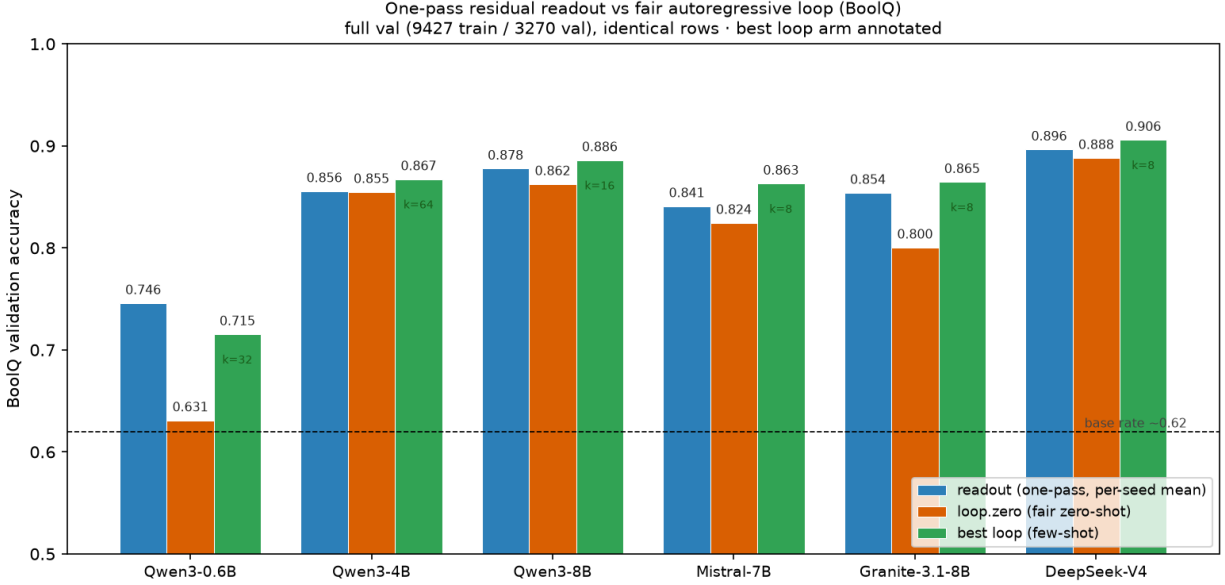


Figure 1: BoolQ: readout vs loop across six models (full val).

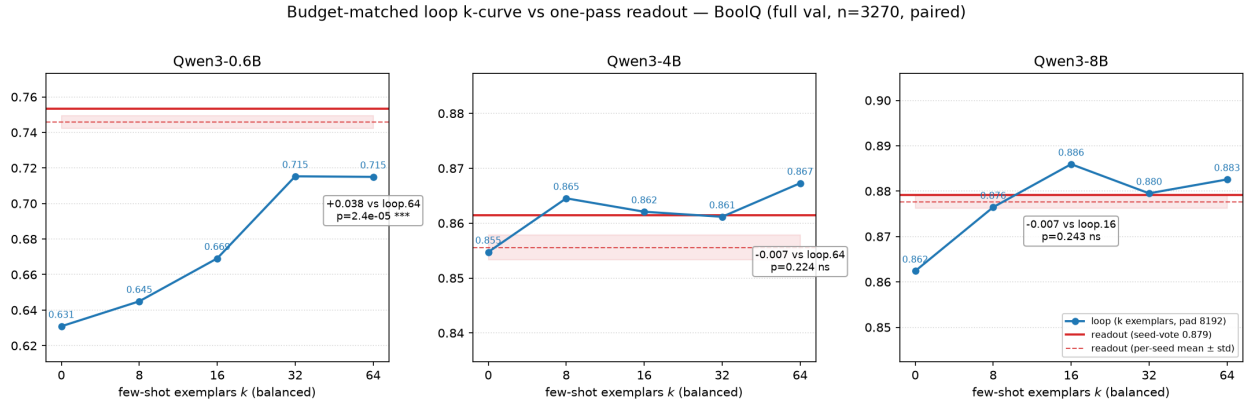


Figure 2: Supervision budget. The loop’s k -curve plateaus by $k=32$ and never passes the readout on BoolQ.

3.3 Readings

1. **Parity or better at five of six models; the edge is largest where the model is weakest.** The readout’s only significant win is at 0.6B — the loop’s verbal channel is the bottleneck there, and the readout channel saturates before it. Mistral-7B is the exception (loop +2.2pt).
2. **The signal is distributed, not localized.** Random-projection heads retain accuracy (perm $\approx$ max $\gg$ noise) at every scale; no privileged subspace carries the verdict (Table 3).
3. **The verdict concentrates in the final token.** Mean-pooled full-context readouts underperform the final-token readout everywhere (full-context $\ll$ final-token) — the answer is computed into the final state, not read off the passage.
4. **The readout is essentially linear.** A multinomial logistic readout tracks the MLP within a few points at every scale (BoolQ 0.6B linear.max 0.750 vs MLP 0.747; 8B 0.874 vs 0.881), so the verdict is a

near-linear read of the residual, not a fragile nonlinear probe. At 0.6B, where the residual does not yet hold the answer (ARC ≈ 0.49 , near chance), no head can invent it — the readout reads what is there.

Context length. The readout reads at `pad_max = 384` against the loop’s 2048–8192. Because the BoolQ gold answer is grounded in the passage, left-truncation could in principle drop long passages asymmetrically; we do not log per-row truncation rates and flag this as a limitation (§7), noting the length-tercile check in §2 shows no length-dependent accuracy collapse.

3.4 Scoring vs decoding

Readout and scoring loop are both one context forward pass, unequal only in context length. A decode loop is a different, more expensive readout. In a small check, greedy decode from an “Answer:” prompt hit the length cap (25/25 at 300 tokens on 0.6B; 7/8 at 1000 tokens on DeepSeek-V4). That is not a latency benchmark.

4 RuleTaker and ARC: serial depth and parametric knowledge

BoolQ is passage-grounded. We test the two ways a task can be harder for a one-pass readout: **serial deduction depth** (RuleTaker) and **parametric knowledge with no passage** (ARC-Challenge).

4.1 RuleTaker 10k/4k: matched loop rows, per-depth strata

RuleTaker [12] is closed-world rule reasoning with labeled deduction depth (0/1/2/3/5 + NatLang), loaded via Hugging Face `tasksources/rulemaker`. Protocol: 10000 train / 4000 val (subsample of the full release), loop on the same 4000 rows, $k \in \{0, 8\}$, batch 8. On Qwen, the matched one-pass MLP **beats the fair loop at every scale** (+4.6 to +9.2pt; all $p < 10^{-8}$; 0.6B prefers zero-shot, 4B/8B prefer 8-shot). The cross-family models sharpen the picture: Mistral-7B +11.1pt ($p = 4.2e-28$), **Granite-3.1-8B** +11.9pt ($p = 1.1e-45$) — **the largest readout margin we measure**. The one clean loop win is DeepSeek-V4 (−1.6pt, $p = 0.029$), driven by the shallow-depth strata.

Per-depth (same rows): no sharp “only the loop works past depth D ” cliff. Accuracy degrades gradually with depth; $d=5$ remains well above chance. The DSV4 loop win concentrates at shallow depths ($d0/d1$, +5–11pt); at 0.6B the loop lags at depth ($d5$ loop.8 0.522 vs. readout 0.625). Depth bins are $n=211$ –1946 ($d3$ dominates) (Fig. 3).

4.2 ARC-Challenge: the loop’s best case

ARC-Challenge [11] is the hard partition of the AI2 Reasoning Challenge (not Chollet’s Abstraction & Reasoning Corpus): grade-school science multiple-choice questions that retrieval and co-occurrence baselines miss. We load `allenai/ai2_arc` config ARC-Challenge and keep 4-choice A–D items only ($\sim 99\%$ of the set). Answers are parametric science knowledge with no passage — they must come from the weights, not the context. On Qwen the loop beats the one-pass classifier at 0.6B (−10.6pt) and 4B (−4.1pt; both $p < 1e-05$). At 8B and DeepSeek-V4 the gap is gone (−0.4pt and −1.0pt, both ns). Cross-family models sit with Qwen at matched readout accuracy, not matched size: Mistral-7B −5.1pt ($p = 9.3e-05$), Granite-8B −8.2pt ($p = 5.8e-10$). Few-shot exemplars teach a task format the residual head must otherwise learn from labels; the gap shrinks as probe accuracy rises.

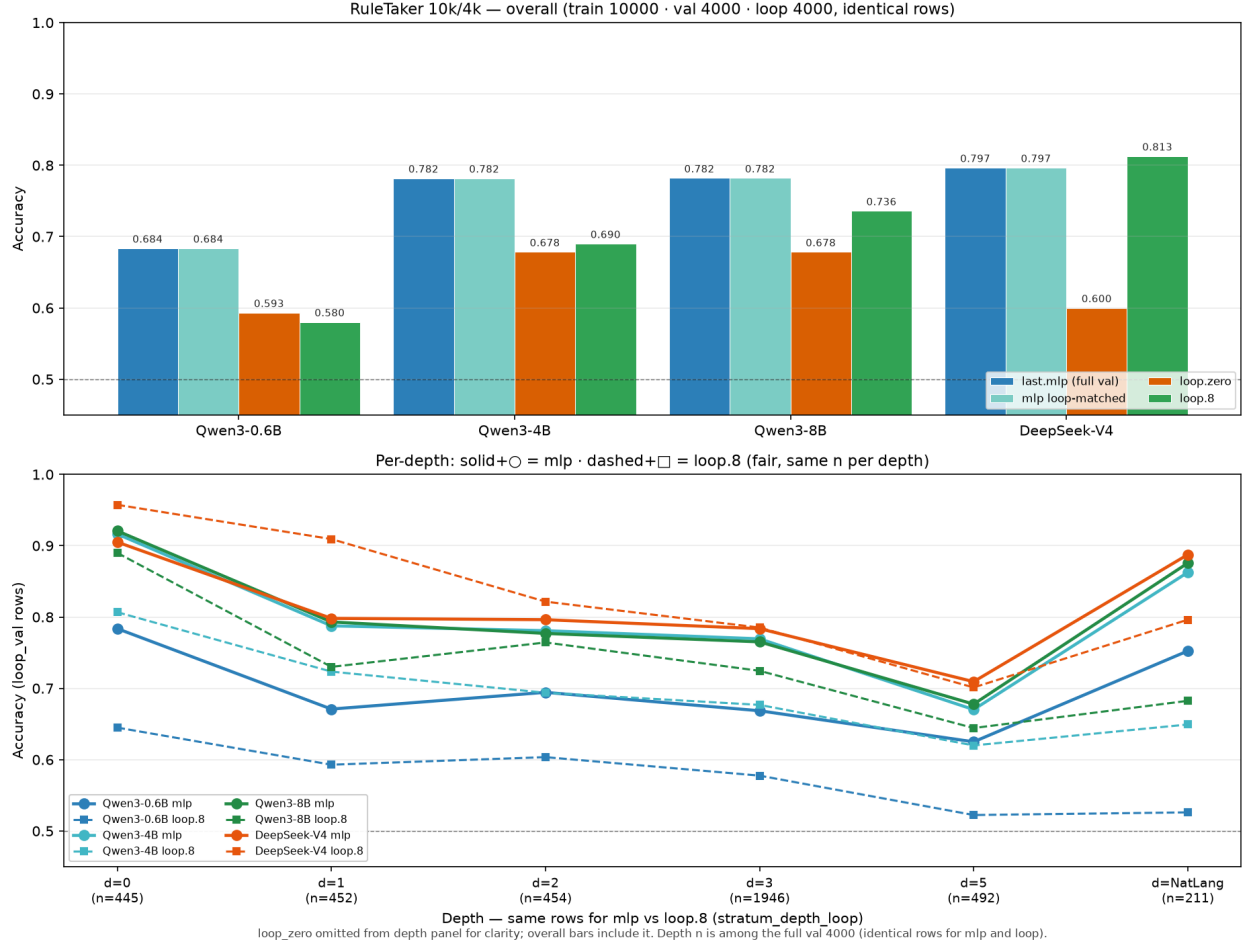


Figure 3: RuleTaker overall + per-depth strata, readout vs loop.

4.3 The task-type regularity

Across 18 task×model cells the organizing fact is capability on ARC, not three special regimes (Fig. 4). **The loop wins on ARC when the one-pass classifier is still weak** (Qwen 0.6B/4B, Mistral, Granite). Elsewhere the one-pass readout wins or ties, with two exceptions: DeepSeek-V4 on RuleTaker and Mistral-7B on BoolQ. Granite/Mistral sit where Qwen did at equal ARC accuracy, not equal size. The Mistral BoolQ cell is a single-model exception we flag rather than explain (§6).

4.4 Readout placement: information across depth

Every headline number above taps the final residual layer. We swept the tap across depth (10 layers, paired McNemar vs the final-layer readout, trained heads persisted at every tap). The curve is the same everywhere: monotone rise, mid-depth plateau, slight terminal decline (Table 2, Fig. 6). A high mid-depth readout accuracy means the residual already contains extractable answer-relevant information; it does not mean the intermediate LM head is correct, or that stopping there reproduces full-model behavior.

- **The final-layer readout is not the best readout.** Mid-depth taps beat the final layer significantly at **11/18 cells and never lose significantly** — the headline numbers were a lower bound on what a gold-label readout can extract, not a claim that later layers add nothing. Across the 18 cells, 10 survive a

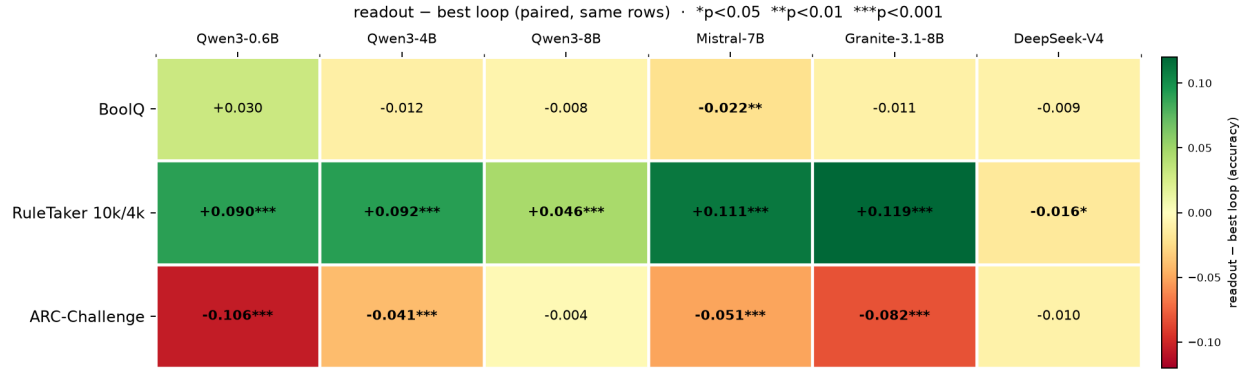


Figure 4: Readout – best loop (paired, same rows) per task×model cell, with McNemar significance. The loop’s significant wins cluster on ARC at low probe accuracy, plus DeepSeek-V4 RuleTaker and Mistral-7B BoolQ.

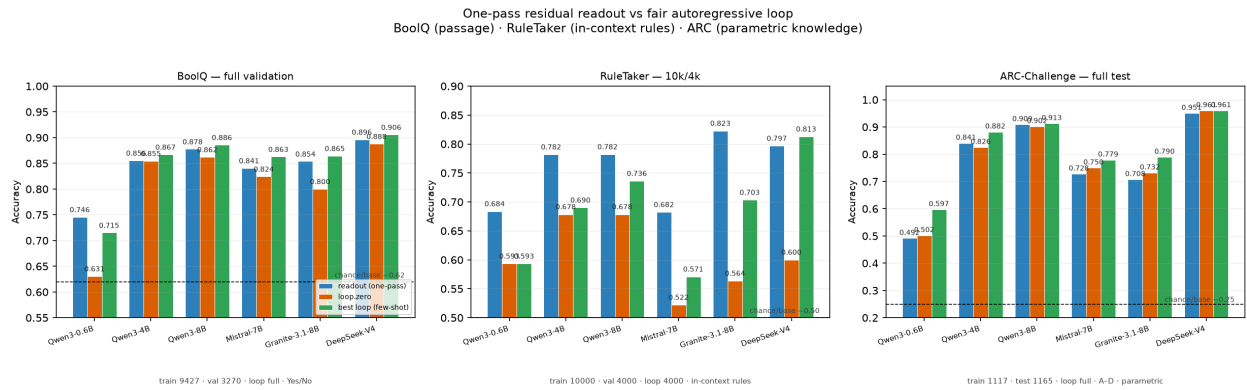


Figure 5: Three-task head-to-head (BoolQ · RuleTaker · ARC) across models.

Benjamini–Hochberg FDR correction at $q < 0.05$ (only BoolQ 4B drops); the “never loses” direction is unchanged.

- **The optimum is task-stable, not scale-driven.** BoolQ/RuleTaker peak at 50–86% depth at every scale; RuleTaker’s Qwen and DeepSeek-V4 optima sit at $\sim 2/3$ depth, while Mistral peaks at mid-stack (L16/32). ARC drifts to the final layer at $\geq 8B$ — once the parametric answer saturates the residual, late-layer specialization stops costing anything.
- **A mid-layer tap can beat the final-layer tap.** Where the loop catches the final-layer readout (BoolQ 8B: L35 ties the loop), the best mid-layer tap beats the final layer (L30 +1.1pt vs L35, $p = 0.016$; Granite L20 +1.4pt vs final, $p = 1e-03$). That is not automatically a win against the loop (BoolQ 8B L30 0.889 vs best loop 0.886).
- **Wide-shallow plateaus earlier.** Mistral (32×4096) onsets at 35–58% depth vs Qwen3-8B’s uniform 69% at the same width — at fixed width, the shallower stack yields a usable label signal sooner. Granite (40×4096 , deep) sits between the two.

Causal hedge. The monotone-rise $\rightarrow$ plateau $\rightarrow$ terminal-decline shape is *consistent with* late-layer specialization toward next-token surface form or confidence correction, but we do not demonstrate it: we never ablated deep layers against generation, never measured calibration vs. correctness by layer, and never probed next-token structure per layer. Three claims should not be conflated: (i) the intermediate LM head

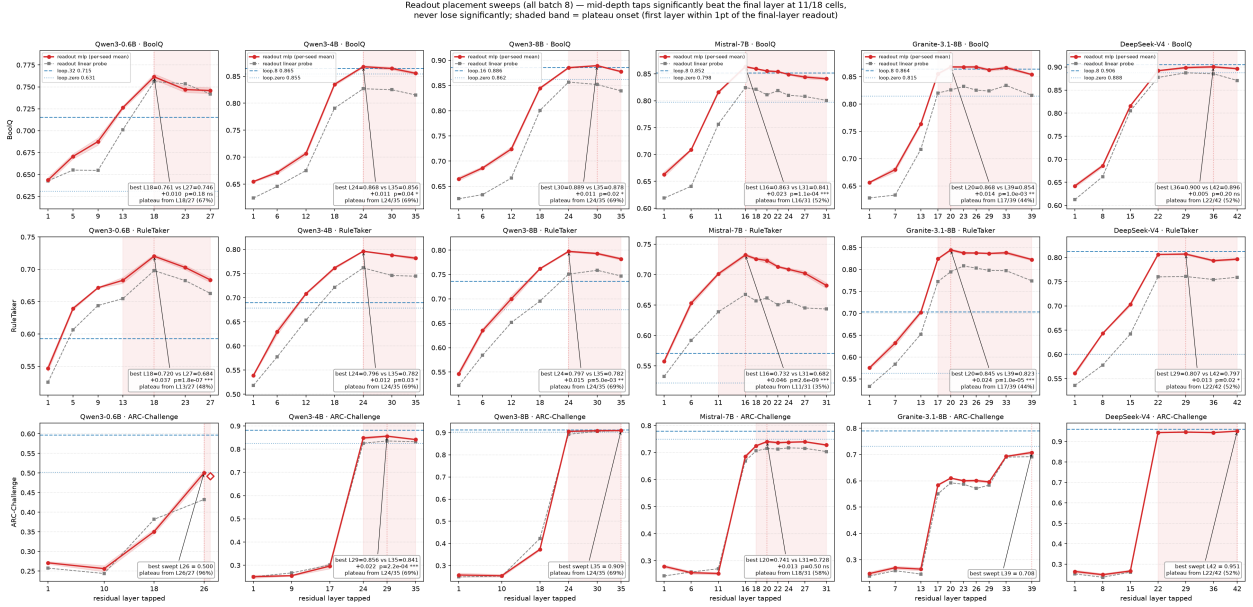


Figure 6: Readout probe-placement sweeps (18 cells). One-pass readout accuracy vs tapped residual layer; loop arms as horizontal references; paired McNemar annotation per panel. Mid-depth taps win significantly at 11/18 cells and never lose.

predicts the correct token; (ii) a small readout can extract the gold answer from h_l ; (iii) stopping at layer l reproduces full-model behavior. The placement sweep addresses (ii). Inferring (iii) from (ii) is the early-exit claim, and it is prior art we do not make (§8).

5 The boundary: where the loop could win, and why it doesn’t cleanly here

A single forward pass is a bounded-depth computation (TC^0 in the circuit-complexity framing); the loop’s advantage should appear exactly where serial depth is irreducible. We hunted that regime and found it only weakly exhibitable on frozen base weights:

- **RuleTaker per-depth** shows a gentle slope, not a regime switch — the loop edges the readout by a few points at $d=5$ (noisy thin bins), and the one decisive loop win (DSV4) concentrates at *shallow* depths, where the frontier model’s in-context deduction is strong.
- **ARC** shows the loop’s real advantage is *parametric retrieval under a learned task format*, not serial compute — and it closes with scale.
- **Synthetic mechanism** (multi-hop transitivity with shuffled facts; Fig. 7): the relation is linearly decodable from the frozen residual (0.93–0.97) where the zero-shot decoder sits at chance; a 2-demo budget surfaces the loop’s inference but never exceeds the one-pass MLP. The measured limit is **order-locked binding**, not depth.

The remaining “loop wins” cell — a TC^0 -hard task on which a *frozen* loop decisively beats one-pass — is largely unoccupied in our sweep; the counter-position (COCONUT, looped-latent) buys serial depth by *training*, which is out of scope here. Because RuleTaker is a 10k/4k subsample of the official release ($d3$ fat), we treat the serial-depth boundary as *indicative*: it shows where a TC^0 gap would appear but does not pin its location on the full test.

The relation is in the residual; the loop never exceeds a one-pass readout

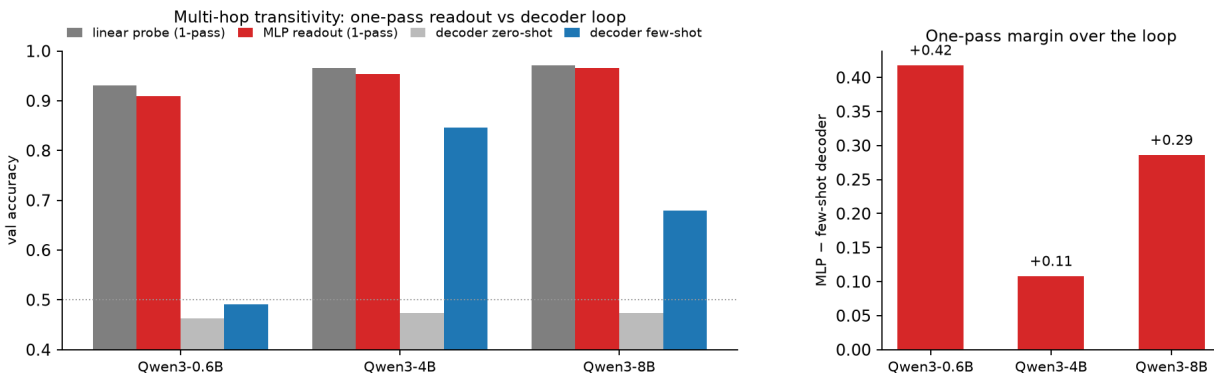


Figure 7: Mechanism (synthetic). Multi-hop transitivity: one-pass readout vs decoder loop across Qwen3 scale. The loop never exceeds a one-pass readout.

6 Discussion

On closed-set classification with a fixed answer set, a single prompt forward pass is enough: a small head on the frozen residual matches a fair autoregressive classifier on most of this task class, so the answer is often already in the vector. The loop still wins on ARC when the probe is weak, and in two other cells (§5).

The residual supports a competitive closed-set classifier. The readout’s largest margins are exactly where the loop is weakest (Granite RuleTaker +11.9pt; 0.6B BoolQ and RuleTaker) — the readout channel saturates before the verbal channel. This matches the internal-knowledge/external-behavior disconnect Orgad et al. [25] document and Gekhman et al. [17] formalize as *hidden knowledge*: a producer readout on the prompt residual can match the classifier the loop provides on this task class. The readout is near-linear (§3): the answer is a near-linear read of a vector the model already computes.

The Mistral BoolQ exception. Mistral-7B is the one model where the loop significantly wins on BoolQ (−2.2pt, $p = 2.6e-03$). It is a single-model, single-task cell, and Mistral’s wide-shallow stack (32×4096) plateaus earliest of the six models (§4.4), so its final-layer residual is the weakest readout substrate relative to its own loop.

Separability of readout from generative interface. The unembedding is one readout of the residual — a fixed, next-token-trained one. That a supervised head matches it is the default expectation once stated that way; the empirical content is *where* it fails (ARC before the residual saturates; DeepSeek-V4 on RuleTaker) and that the failure modes are orderly.

Specialist frontier (acknowledged). Fine-tuned encoder classifiers (RoBERTa-large ~ 0.87 , DeBERTa-v3 ~ 0.88 , T5-11B 0.91 on BoolQ) **beat** our 8B one-pass readout (0.878). Our claim is readout-vs-loop on a frozen backbone — the comparison a serving stack faces when the weights are already loaded — not readout-vs-fine-tuned-specialist. Where a specialist can be trained and shipped, it wins on raw accuracy; the readout’s economics are marginal-cost (the forward pass already happened).

What we are not claiming. A frozen continuous latent loop works (it drifts); the loop never computes; we beat fine-tuned specialists; we shipped residual-first inference; we introduce early exit or adaptive computation; intermediate confidence is a stop rule; later layers can be skipped because a mid-depth readout is accurate. Paper 2 explores the serving architecture as a proposal.

7 Limitations

- **RuleTaker is a 10k/4k subsample** of the official release (natural depth mix, $d3$ fat), not the full test; per-depth bins are $n=211-1946$.
- **Multiple testing:** Table 1’s 18 McNemar p -values are reported uncorrected (stars); all 12 raw-significant cells survive the Benjamini–Hochberg FDR correction at $q<0.05$. Placement is stricter: 1 of 11 raw mid-depth wins drops under the same correction (10/18 remain).
- **Truncation is not logged per row:** the readout reads at $\text{pad_max} = 384$ (left-truncated) against the loop’s 2048–8192, and we do not record the fraction of rows truncated, so a length-asymmetric truncation cost on long-passage rows cannot be ruled out from the logged data (the per-length deconfound shows no accuracy collapse, but it is a coarse check).
- **Probe vs few-shot scorer:** the readout is fit on the training split; the loop is in-context scoring, not a fine-tuned decoder. On BoolQ the loop’s k -curve plateaus by $k=32$ and never passes the readout where the readout was winning.
- **Reproducibility:** per-layer heads and per-row predictions are persisted; the two oldest headline runs predate head persistence. MoE active-param FLOPs are unmeasured.
- **Scope:** closed-set classification with a fixed answer set (Yes/No, A–D). The loop baseline is a scoring classifier, not chain-of-thought or free-form generation. No claim transfers to open-ended decoding; the finding is that one prompt forward pass *suffices* for this class, not that every task is one-pass.

8 Related Work

Accessing what a frozen model knows. *Behavioral* work [29] shows CoT rationales can be unfaithful — text-only access, no representation. *Latent-belief* work reads the representation directly: CCS [5] finds an unsupervised truth direction (belief vs. gold-label answer recovery), pre-CoT belief probes [13], belief-formation timing [3], and the correctness/self-knowledge cluster [6, 17, 22, 25]. Orgad et al. detect errors (and, in their §6, re-rank 30 samples); Gekhman et al. formalize *hidden knowledge* and re-rank up to 1k samples with a verifier probe on $h(q, a)$ — both score *candidates the model already emitted*, not a producer head on the prompt residual alone, and neither holds that head to a fairly-conditioned closed-set loop. *Decoding-suboptimality* work [1, 4, 8, 9] and, closest, Hazenoot et al. [20] show hidden-state readouts beat the token-probability decision rule or zero-shot prompting — baselines their own analyses show are weak. *Probe-for-verification* work [23] uses internal states to improve generation, not replace it.

Residual-stream lenses and representation tools ask *what* a layer already encodes, not whether a small gold-label readout matches a fair loop as a classifier. The logit lens [24] and tuned lens [2] decode intermediate residuals toward the vocabulary; Patchscopes [18] and representation engineering [35] intervene on or read directions for interpretability and control. We measure gold-answer accuracy of a frozen-residual readout vs. the same model’s few-shot scorer.

COCONUT [19] and looped-latent transformers [26] *train* a continuous latent loop to buy serial depth — they set the boundary we test without training. Early-exit and adaptive computation are established [7, 15, 16, 27, 28, 31, 33, 34]: they decide *when to stop* from intermediate predictions or confidence. We measure how much gold-label information a small readout can extract from intermediate residuals — representation sufficiency, not a license to skip later layers. Intermediate confidence is not automatically a reliable stop rule [32, 34]; later layers may refine calibration after a decision is already linearly readable [21]; and early-exit opportunity is not universal in modern LLMs, especially MoE models [30]. Trajectory-dynamics probes [14]

read layer-wise displacement across all tokens for OOD monitoring — a richer readout than our static one, aimed at a different goal.

9 Conclusion

On closed-set classification with a fixed answer set, a single prompt forward pass is enough: a small readout head fit to the frozen residual recovers the answer without generating tokens — and matches the strongest *closed-set scoring* use of the autoregressive loop on BoolQ and RuleTaker, except two cells. On ARC the loop wins when the one-pass classifier is still weak (Qwen $-10.6 \rightarrow -0.4\text{pt}$; Granite-8B still -8.2pt). The remaining BoolQ cells are statistical ties (four of six within $\pm 1.2\text{pt}$). The readout is near-linear: the answer is a near-linear read of a vector the model already computes.

Data availability

Dataset releases used as-is. BoolQ [10] (google/boolq). RuleTaker [12] (tasksource/ruletaker), 10k/4k subsample. ARC-Challenge [11] (allenai/ai2_arc, config ARC-Challenge).

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Table 1: One-pass residual readout vs. fair AR closed-set scorer (accuracy). Readout: MLP on frozen final residual (*per-seed mean*; BoolQ/ARC finals match Table 2; RuleTaker is the 10k/4k run). Loop: next-token log-prob over the closed answer set, full context, balanced few-shot — not free-form generation. loop.0 / loop.8 = zero- / 8-shot; *best loop* = stronger arm (k in parentheses; RT 0.6B and ARC DSV4 prefer zero-shot). Δ = readout – best loop (best-of makes Δ conservative). Paired McNemar on identical rows using 4-seed vote preds (mean-vote ≤ 0.9 pt; never flips sig/ns). Stars mark *uncorrected* p (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$); all 12 raw-significant cells remain significant under Benjamini–Hochberg FDR at $q < 0.05$ across the 18 cells. ^aBoolQ val 3270; Qwen $k \leq 64$ pad 8192, else $k=8$ pad 2048. ^bRuleTaker 10k/4k, val 4000 (same rows for readout and loop). Table 2 uses the same 10k/4k protocol; displayed RuleTaker finals match. ^cARC-Challenge test 1165, $k=8$.

Task	Model	Readout	loop.0	loop.8	best loop	Δ (readout – loop)	p (McNemar)
BoolQ ^a	Qwen3-0.6B	0.746	0.631	0.645	0.715 (32)	+0.030 ***	3.2e−05
	Qwen3-4B	0.856	0.855	0.865	0.867 (64)	−0.012	0.094
	Qwen3-8B	0.878	0.862	0.876	0.886 (16)	−0.008	0.243
	Mistral-7B	0.841	0.824	0.863	0.863 (8)	−0.022**	0.003
	Granite-3.1-8B	0.854	0.800	0.865	0.865 (8)	−0.011	0.076
	DeepSeek-V4-Flash	0.896	0.888	0.906	0.906 (8)	−0.009	0.078
RuleTaker ^b	Qwen3-0.6B	0.684	0.593	0.580	0.593 (zero)	+0.090 ***	1.0e−20
	Qwen3-4B	0.782	0.678	0.690	0.690 (8)	+0.092 ***	6.8e−29
	Qwen3-8B	0.782	0.678	0.736	0.736 (8)	+0.046 ***	1.6e−09
	Mistral-7B	0.682	0.522	0.571	0.571 (8)	+0.111 ***	4.2e−28
	Granite-3.1-8B	0.823	0.564	0.703	0.703 (8)	+0.119 ***	1.1e−45
	DeepSeek-V4-Flash	0.797	0.600	0.813	0.813 (8)	−0.016*	0.029
ARC-Challenge ^c	Qwen3-0.6B	0.492	0.502	0.597	0.597 (8)	−0.106***	3.2e−11
	Qwen3-4B	0.841	0.826	0.882	0.882 (8)	−0.041***	2.9e−06
	Qwen3-8B	0.909	0.902	0.913	0.913 (8)	−0.004	0.567
	Mistral-7B	0.728	0.750	0.779	0.779 (8)	−0.051***	9.3e−05
	Granite-3.1-8B	0.708	0.732	0.790	0.790 (8)	−0.082***	5.8e−10
	DeepSeek-V4-Flash	0.951	0.961	0.959	0.961 (zero)	−0.010	0.064

Table 2: Readout probe-placement sweeps (18 cells, all batch 8): best mid-depth tap vs the final-layer readout. Δ = displayed best – displayed final (per-seed means). p is paired McNemar on 4-seed vote (mean-vote can differ by ~ 1 pt). RuleTaker is the same 10k/4k protocol as Table 1. *Layers* = stack depth (28/36/36/32/40/43). *Plateau onset* = first swept layer within 1 pt of the final-layer readout — a lightweight readout extracts nearly as much label information from there on as from the final residual; at DeepSeek-V4 this happens by L22/42 (52% depth) on *all three tasks*, with ARC showing the sharpest snap-in (near-chance at L15 $\rightarrow$ 0.944 at L22). This is representation sufficiency, not a claim that later layers can be skipped. Mid-depth taps win significantly at 11/18 cells raw (10/18 after Benjamini–Hochberg FDR at $q < 0.05$) and never lose significantly.

Task	Model	#	Tap	Dep.	Plateau	best	final	Δ	p
BoolQ	Qwen3-0.6B	28	L18	64%	L18 (67%)	0.761	0.746	+0.015	0.18
	Qwen3-4B	36	L24	67%	L24 (69%)	0.868	0.856	+0.012*	0.038
	Qwen3-8B	36	L30	83%	L24 (69%)	0.889	0.878	+0.011*	0.016
	Mistral-7B	32	L16	50%	L16 (52%)	0.863	0.841	+0.022***	1.1e−04
	Granite-3.1-8B	40	L20	51%	L17 (44%)	0.868	0.854	+0.014**	1.0e−03
	DeepSeek-V4-Flash	43	L36	86%	L22 (52%)	0.900	0.896	+0.004	0.20
RuleTaker	Qwen3-0.6B	28	L18	64%	L13 (48%)	0.720	0.684	+0.036***	1.8e−07
	Qwen3-4B	36	L24	67%	L24 (69%)	0.796	0.782	+0.014*	0.027
	Qwen3-8B	36	L24	67%	L24 (69%)	0.797	0.782	+0.015**	5.0e−03
	Mistral-7B	32	L16	50%	L11 (35%)	0.732	0.682	+0.050***	2.6e−09
	Granite-3.1-8B	40	L20	51%	L17 (44%)	0.845	0.823	+0.022***	1.0e−05
	DeepSeek-V4-Flash	43	L29	67%	L22 (52%)	0.807	0.797	+0.010*	0.020
ARC-Challenge	Qwen3-0.6B	28	L26	93%	L26 (96%)	0.500	0.492	+0.008	0.082
	Qwen3-4B	36	L29	81%	L24 (69%)	0.856	0.841	+0.015***	2.2e−04
	Qwen3-8B	36	L35	97%	L24 (69%)	0.909	0.909	0.000	1.00
	Mistral-7B	32	L20	63%	L18 (58%)	0.741	0.728	+0.013	0.50
	Granite-3.1-8B	40	L39	100%	L39 (100%)	0.708	0.708	0.000	—
	DeepSeek-V4-Flash	43	L42	98%	L22 (52%)	0.951	0.951	0.000	—

Table 3: Controls, all 18 cells. Every headline number clears all four controls. *label-shuffle*: the head refit on permuted labels sits at chance. *random-projection*: a head on random Gaussian projections retains accuracy with permuted-basis $\approx$ max-basis $\gg$ noise floor — the signal is *diffuse*. *full-context*: a mean-pooled full-context readout underperforms the final-token readout — the verdict is computed into the final state, not read off the passage. Balanced chance is 0.50 (BoolQ, RuleTaker) and 0.25 (4-way ARC); the *majority-class* baseline is $\approx$ 0.62 (BoolQ), 0.50 (RuleTaker), 0.27 (ARC).

Task	Model	shuffle	rp max	rp perm	rp noise	full-ctx	final
BoolQ	Qwen3-0.6B	0.533	0.734	0.725	0.601	0.667	0.746
	Qwen3-4B	0.542	0.861	0.859	0.604	0.717	0.856
	Qwen3-8B	0.528	0.874	0.874	0.607	0.739	0.878
	Mistral-7B	0.537	0.810	0.803	0.607	0.685	0.841
	Granite-3.1-8B	0.532	0.848	0.849	0.607	0.679	0.854
	DeepSeek-V4	0.522	0.889	0.891	0.602	0.678	0.896
RuleTaker	Qwen3-0.6B	0.498	0.657	0.663	0.502	0.532	0.684
	Qwen3-4B	0.516	0.748	0.752	0.490	0.619	0.782
	Qwen3-8B	0.506	0.760	0.755	0.502	0.634	0.782
	Mistral-7B	0.503	0.656	0.646	0.502	0.521	0.682
	Granite-3.1-8B	0.504	0.792	0.785	0.502	0.599	0.823
	DeepSeek-V4	0.509	0.768	0.759	0.503	0.526	0.797
ARC	Qwen3-0.6B	0.278	0.470	0.470	0.254	0.278	0.492
	Qwen3-4B	0.269	0.802	0.803	0.255	0.349	0.841
	Qwen3-8B	0.287	0.869	0.867	0.251	0.399	0.909
	Mistral-7B	0.259	0.679	0.678	0.251	0.277	0.728
	Granite-3.1-8B	0.284	0.598	0.588	0.251	0.279	0.708
	DeepSeek-V4	0.269	0.942	0.940	0.236	0.439	0.951

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